

PAPER

Fast and accurate robotic optical detection of exfoliated graphene and hexagonal boron nitride by deep neural networks

To cite this article: Young Jae Shin *et al* 2021 *2D Mater.* **8** 035017

View the [article online](#) for updates and enhancements.

You may also like

- [Deep neural network-assisted radio frequency inductively coupled plasma discharge simulation](#)
Yang Zhao, Wenyi Chen, Zongcheng Miao et al.
- [Training improvements for ultrasound beamforming with deep neural networks](#)
A C Luchies and B C Byram
- [Deep learning-assisted magnetized inductively coupled plasma discharge modeling](#)
Yang Zhao, Wenyi Chen, Zongcheng Miao et al.



PAPER

Fast and accurate robotic optical detection of exfoliated graphene and hexagonal boron nitride by deep neural networks

Young Jae Shin^{1,2} , Wheemyung Shin³, Takashi Taniguchi⁴, Kenji Watanabe⁴ , Philip Kim¹ and Sung-Ho Bae³ ¹ Department of Physics, Harvard University, Cambridge, MA 02138, United States of America² Center for Functional Nanomaterials, Brookhaven National Laboratory, Upton, NY 11973, United States of America³ Department of Computer Science and Engineering, Kyung Hee University, Yongin, Gyeonggi-do 17104, Republic of Korea⁴ National Institute for Materials Science, 1-1 Namiki, Tsukuba 305-0044, JapanE-mail: yshin@bnl.gov and shbae@khu.ac.kr**Keywords:** graphene, hBN, deep neural network, automatic optical detectionSupplementary material for this article is available [online](#)

RECEIVED

15 November 2020

REVISED

20 December 2020

ACCEPTED FOR PUBLICATION

29 December 2020

PUBLISHED

20 April 2021

Abstract

Van der Waals (vdW) heterostructure constructed from atomically thin layered materials provides quantum material platforms with emergent physical phenomena and novel device applications. While stacking atomically thin vdW layers in combination with automate machine vision identification and semi-automated stacking have been demonstrated, a combination of machine learning based automatic detection/identification assembly capability is necessary to further advance vdW heterostructure fabrication. Here, we developed a new automatic optical detection technique with a deep neural network (DNN) incorporated into a motorized microscope that automatically scans entire silicon wafers to detect and identify two-dimensional (2D) materials. We demonstrated the automated combination on an optical microscope (OM) with a DNN algorithm that enables identification and classification of graphene with different sizes, shapes and thicknesses. For this purpose, we trained a representative DNN for object detection with approximately 1000 OM images, resulting in high accuracy in detection and classification. We further verified the effectiveness of graphene trained DNN (GT-DNN) in practice, by confirming the yield of flakes depending on exfoliation method. In addition, we also showed the transferability of our method to other 2D materials. Our pre-trained GT-DNN was used to train a detection model for hexagonal boron nitride with only a few samples, resulting in further improvement of the accuracy of DNN-based detection. Our experimental method can effectively be extended to automatic detection and assembly of a plethora of 2D materials.

1. Introduction

Building van der Waals (vdW) heterostructures using atomically thin crystallites via mechanical manipulation is a widely accepted experimental technique for fabricating two-dimensional (2D) quantum structures. At vdW interfaces, a vast array of emergent physical phenomena occur, creating promising opportunities for quantum information science applications [1, 2]. Most frequently, 2D materials are obtained in bulk crystal form where layers are weakly connected with vdW forces and therefore can be exfoliated mechanically [3, 4]. There are now more than 2000 recognized vdW materials that can, in principle,

be exfoliated down to the atomically thin crystallite limit [5, 6]. Those atomically thin 2D materials can be restacked into quasi three-dimensional (3D) structures by vdW stacking [7, 8]. Unlike the chemical synthesis approach or molecular beam epitaxy, the vdW layering approach is free from constraints on chemical compatibility between adjacent layers and atomic-scale interdiffusion, making it feasible to create architectures with atomically sharp interfaces between layers [9]. vdW stacking also allows control of stacking angles, enabling moiré superlattice engineering, which recently led the discovery of unconventional superconductivity in graphene bilayers [10, 11].

Despite the promise of stacked 2D heterostructures, building state-of-art vdW heterostructures demands multiple levels of painstaking manual operations that require experience, dexterity and time. A single high-quality multiple layer heterostructures can take more than a day to complete. It is virtually impossible to fully exploit the enormous combinatorial material opportunity provided by this vdW system without a breakthrough in methods for synthesizing stacked heterostructures.

Currently, one of the bottlenecks to assembling vdW heterostructures is finding target flakes for preparation of stacking. The mechanical exfoliation process that accompanies 2D flake scanning under an optical microscope (OM) is the most commonly employed method for extracting high-quality 2D flakes in laboratories [12]. Typically, an OM is used for detecting and recognizing graphene and other 2D material flakes on 285 nm SiO₂. The contrast between graphene and SiO₂ is a crucial factor in the optical detection of graphene flakes. However, often the contrast ratio difference between graphene with different thicknesses is small. Several approaches using a machine learning algorithm for image recognition have been introduced [13]. In particular, a neural-network-based algorithm was recently introduced in image analysis and thickness identification of exfoliated 2D materials. As long as a large training sets are provided, this neural-network based algorithm can provide high prediction accuracy that can overcome 2D crystal variation, microscope configurations and imaging conditions.

In this work, we demonstrate our proposed method, a graphene trained deep neural network (GT-DNN), which is an automated process of imaging and flake recognition using a DNN. We apply our GT-DNN to mechanically exfoliated graphene and hexagonal boron nitride (hBN) on a wafer scale. Since there have been significant advances in object detection with DNNs [14–19], we used a representative DNN architecture for object detection, RetinaNet [14] in order to detect materials. Previous research [20, 21] has shown that DNNs can be used to identify and segment flakes. However, the data sets used for training these DNNs relied on tagged and optimized images from previous research where the target flakes were already highlighted and located at the center region of the images by experienced researchers; thus, the trained models show limited localization ability on a small area (see figures S1 (graphene) and S2 (hBN)) (available online at stacks.iop.org/2DM/8/035017/mmedia). Moreover, the DNN models were applied on a fixed system where limited microscopy conditions, such as color space and white balance, were used for acquiring training image samples, resulting in low generalization performance of the DNN models on test samples taken from various microscopy conditions

To apply our GT-DNN for automated image acquisition and detection under various microscopy conditions, it is necessary that the program should be able to handle photographic images with diverse color space and white balance (see figure S3 for example). It is therefore important to train a highly generalized detection model to diverse microscopy conditions.

The GT-DNN model presented in our work also enabled a fully automated robotic detection system. We successfully combined the GT-DNN model with a motorized microscope and developed an automatic graphene detection system that scans and detects graphene. The GT-DNN can also classify target flake thickness (mono-, bi- and tri-layer). Transferability to other materials is another important consideration. In particular, for extremely rare materials, gathering enough training data is not possible. With this transfer learning technique, we transferred the graphene detection system to hBN samples with a small amount of training data. This experiment demonstrates that the GT-DNN's potential to be expanded to other 2D material flakes.

Figure 1 presents the flowchart of the proposed GT-DNN. After a substrate chip (1.2 cm × 1.2 cm) with a mechanically exfoliated crystal is loaded, the motorized microscope starts scanning the substrate as in figure 1 (left). Each photograph generated by scanned optical imaging is saved with its shift offset so that every image can be concatenated to create a picture of the whole substrate. These images are used as input of GT-DNN and the output is the predicted locations and labels of the materials with their confidences. The location of each flake is presented as a bounding box that contains a detected flake sample. We adopted RetinaNet [14] as the baseline model. After the GT-DNN predicts all the bounding boxes for each image, the accumulated output box information constructs a library for the whole scanned substrate.

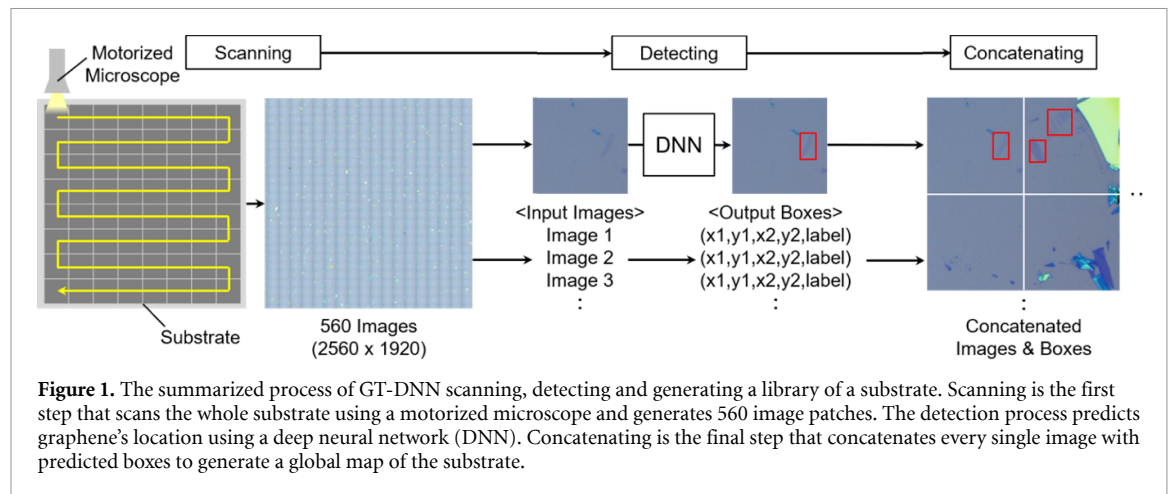
2. Experimental

2.1. Cleavage of graphene and hBN

Graphene and hBN were cleaved on 285 nm and 90 nm Si oxide thermally grown on standard 4" Si wafers. Natural graphite (NG) crystals (NaturGrafit GmbH) and hBN bulk crystals (HQgraphene) were mechanically exfoliated with Nitto Denko thermal releasing tape (TRT). The silicon oxide substrate was treated in oxygen plasma for 10 min (PlasmaEtch PE-50, 300 mbar O₂, 100 W) and the 2D material-loaded tape was immediately applied to the silicon oxide surface. The tape was subsequently released from the surface by heating to 100 °C on a hot plate.

2.2. Motorized Nikon microscope setting

Silicon oxide wafers were diced to 1.2 cm × 1.2 cm. The microscope scanned whole chips as shown in figure 1. Focus was recalibrated automatically every



three shots. White balance and color space were fixed throughout scanning. Chip carriers could hold nine chips and the microscope was capable of scanning nine chips.

2.3. Training the DNN

The DNN architecture we used is a representative DNN model for object detection, i.e. RetinaNet [14]. Our RetinaNet's backbone model was ResNet50 [22] and other detailed settings, such as anchors, basically followed the original paper if not specified. For all experiments, we used Adam optimizer [23] and our target loss was cross entropy (CE) loss and focal loss [14] was not used, as CE Loss yielded better performance in our task (figure S4 of supplementary information). We also adjusted the parameters of focal loss (α and γ), but no setting was better than using only CE Loss. By default, we initialized weight from ImageNet pretraining because it shortens training time [24]. We started training with a 5×10^{-4} learning rate and divided it by ten when the training loss did not decrease, stopping after 50 K iterations. Batch size was four. For data augmentation [25] and class balancing, we randomly cropped training images several times with 1152×1152 sized images if and only if there was at least one target material (box) in the sample. After cropping, we conducted random horizontal and vertical flipping with 50% chance, $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ rotations with the equal probability and additional random deformations. In addition, we modulated color distributions and brightness of the original images randomly. We conducted experiments under two V100 GPUs with the CUDA 10.1 library. The inference time was about 0.87 s per image (2560×1920) on a single V100 GPU.

2.4. Data availability

Our source code was built with the Keras [26] deep learning library. The reason we chose Keras [26] is for simplicity. However, Pytorch [27] or Tensorflow [28] are also available for the proposed method. For

reproducibility, our source code and whole dataset will be made available online soon.

3. Results and discussion

3.1. Dataset preparation

All graphene and hBN used in this paper were exfoliated on top of 285 nm or 90 nm SiO_2/Si substrates. A total of 1134 OM images containing exfoliated flakes of graphite (1006 images) and hBN (128 images) were collected and labeled for training and testing the DNN. Each label consists of the target flake's layer information and size/location in rectangular box form. For the graphene dataset, 679 images were obtained by ten human researchers in 2752×2192 pixel resolution with four different camera magnifications $\{\times 10, \times 20, \times 50, \times 100\}$ (Set G_{human} , where G stands for graphene) and 327 images were auto-scanned with motorized microscope at 2560×1920 pixel resolution with a $\times 20$ in-camera magnification (Set G_{auto}). On the other hand, all hBN images were auto-scanned with motorized microscope at 2560×1920 pixel resolution with a $\times 20$ in-camera magnification (Set hBN_{auto}). The motorized microscope used in this study was a Nikon LV100ND (Tokyo, Japan).

In the graphene dataset, we first divide available images into two categories: a train-validation set and a test set. For Set G_{human} , we use 554 images for the train-validation set and 125 images for the test set. For Set G_{auto} , 297 and 30 images are used for train-validation and test sets, respectively. For Set hBN_{auto} , 108 and 20 images were used for train-validation and test sets, respectively. Furthermore, our GT-DNN classifies the detected graphene (or hBN) into mono-, bi- and tri-layers for Set G_{auto} and Set hBN_{auto} .

3.2. Metric

To evaluate the proposed GT-DNN, we use average precision (AP) as a quantitative metric that is generally used in the object detection area [29]. The

AP is derived from precision and recall, which are defined as

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP});$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}).$$

where TP, FP and FN stand for true positive, false positive and false negative samples, respectively. The AP is then expressed as

$$\text{AP} = \int_1^0 p(r) dr$$

where $p(r)$ is precision function of recall value (r). Therefore, AP indicates the total underlying area covered by $p(r)$ curve in the precision (y -axis)-recall (x -axis) domain.

To judge which detected box is correct, we set the confidence (output value for the target label) threshold to be 0.5 [29] and intersection over union (IoU) threshold to be 0.3 by default, where IoU is defined as

$$\text{IoU} = \frac{\text{Target box} \cap \text{Predicted box}}{\text{Target box} \cup \text{Predicted box}}.$$

The IoU threshold is generally set to be 0.5 in the literature [29]. However, we found that many predicted boxes under 0.5 in IoU also show reasonable localization. As the distribution of graphene flakes is sparse in the whole substrate, its yield (proportional to Recall) is more important to consider than the Precision for graphene flakes. Therefore, to increase the yield of graphene, we set the IoU threshold to 0.3.

3.3. Visual analysis on behavior of the DNN

In general, the shape and contrast of graphene flakes in OM images are rarely distinguishable from the background substrates, which makes detecting graphene flakes challenging. To make flakes in OM images more distinguishable, researchers adjust the filter parameters of the OM. Likewise, convolution layers in GT-DNN extract the distinguishable features from graphene flake and project them into the space having higher distinguishability in a progressive manner.

Figure 2(a) visualizes features from the original input image and the six output features from the 1st- to 6th-level convolution layers using t -distributed stochastic neighbor embedding (t -SNE) [30]. The t -SNE [30] serves to cluster features with high similarity and projects the clustered features to a 2D space for visualization purpose. In the t -SNE maps, the red, blue, cyan and yellow colored dots indicate the features of background, mono-layer graphene, bi-layer graphene and tri-layer graphene flakes, respectively.

Investigating the progress of t -SNE [30] distribution, we find that the convolution layers tend to extract distinguishable features of various-layered

graphene flakes step by step. Specifically, the 1st convolution layer starts briefly dividing the graphene flakes and background features as shown in the feature distribution of two large clusters. The following convolution layers (2–6) perform more elaborate clustering tasks, i.e. they divide the whole cluster of graphene flakes into the mono-, bi- and tri-layered graphene flakes. For further analysis of the internal behavior of the GT-DNN, we apply Grad-CAM [31], which is a representative tool used to interpret DNN's behavior. That is, Grad-CAM [31] performs inverse mapping from the predicted output to the regions of the input image that contribute most to the given output. Figure 3 shows the visual output (heat map) of Grad-CAM for the GT-DNN. As shown in figure 3, the GT-DNN focuses on regions of graphene flakes, producing the highest activation on the graphene regions. These results imply that our GT-DNN is optimized to detect graphene flakes, i.e. the GT-DNN amplifies signals from graphene and reduces the impact of noises, dust and illumination so that they can be easily classified.

3.4. Verification of GT-DNN generalization performance

In order to verify the generalization performance of the GT-DNN, we train and test the GT-DNN with OM images for graphene where various trained researchers take OM images (Set G_{human}). Each human researcher tends to have a different strategy for taking OM images where ambient luminance, white balance, zoom ratio and contrast ratio values may vary.

Figure 4(a) shows some examples for the aforementioned data variations. The green boxes in the figures are labeled by researchers and the red boxes are predicted by the GT-DNN. In summary, our GT-DNN achieved AP 90.41% with Precision 86.47% and Recall 91.87% under the default setting (figure 4(b)). With these high AP, precision and recall results, our GT-DNN succeeds in detecting graphene, which was not achieved previously without strict environmental control. Our experimental results clearly show that the proposed GT-DNN is highly generalized to variable environmental setups and can be adapted to other systems, such as fully automated detection with a motorized microscope, which will be described in the next section. In addition, to the best of our knowledge, it is the first experiment that revealed the Recall rate of a graphene detection method.

3.5. Proposed fully automated detection with a motorized microscope

Here we will describe the proposed fully automated robotic graphene detection system. To further automatize the preceding GT-DNN, we used auto-scanned images obtained from a motorized microscope system (Set G_{auto}) to train our GT-DNN. Set G_{auto} has a relatively constant background color and

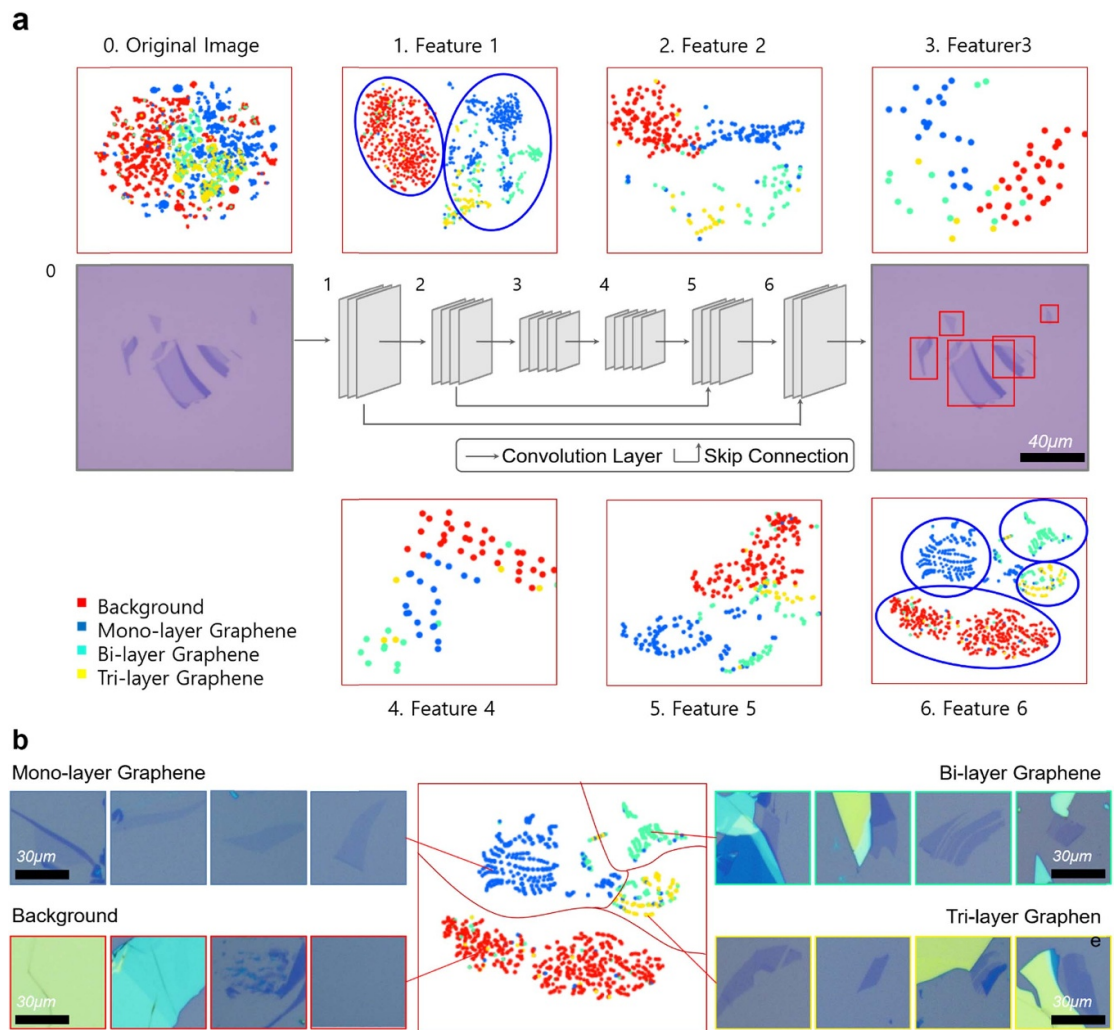


Figure 2. Feature visualization using *t*-SNE [30]. (a) Visualization of the features clustered with high similarity and projection of clustered features to a 2D space where the red, blue, cyan and yellow colored dots indicate the features of background, mono-layer graphene, bi-layer graphene and tri-layer graphene flakes, respectively. (b) Decision boundary and example cases from each cluster.

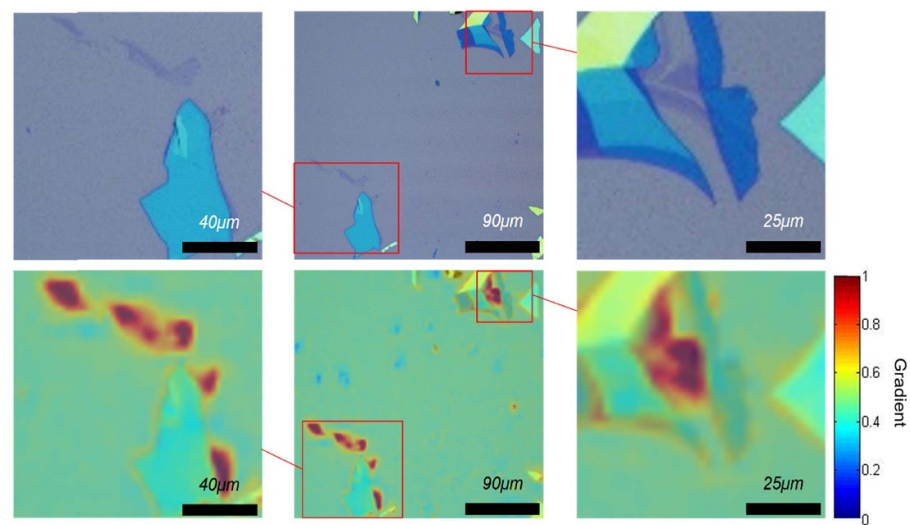


Figure 3. The heat map generated from Grad-CAM [31]. Grad-CAM [31] is an inverse mapping tool to spot the contributing regions of input images to predict the output. The generated heatmap shows that our GT-DNN focuses on graphene areas to make a decision.

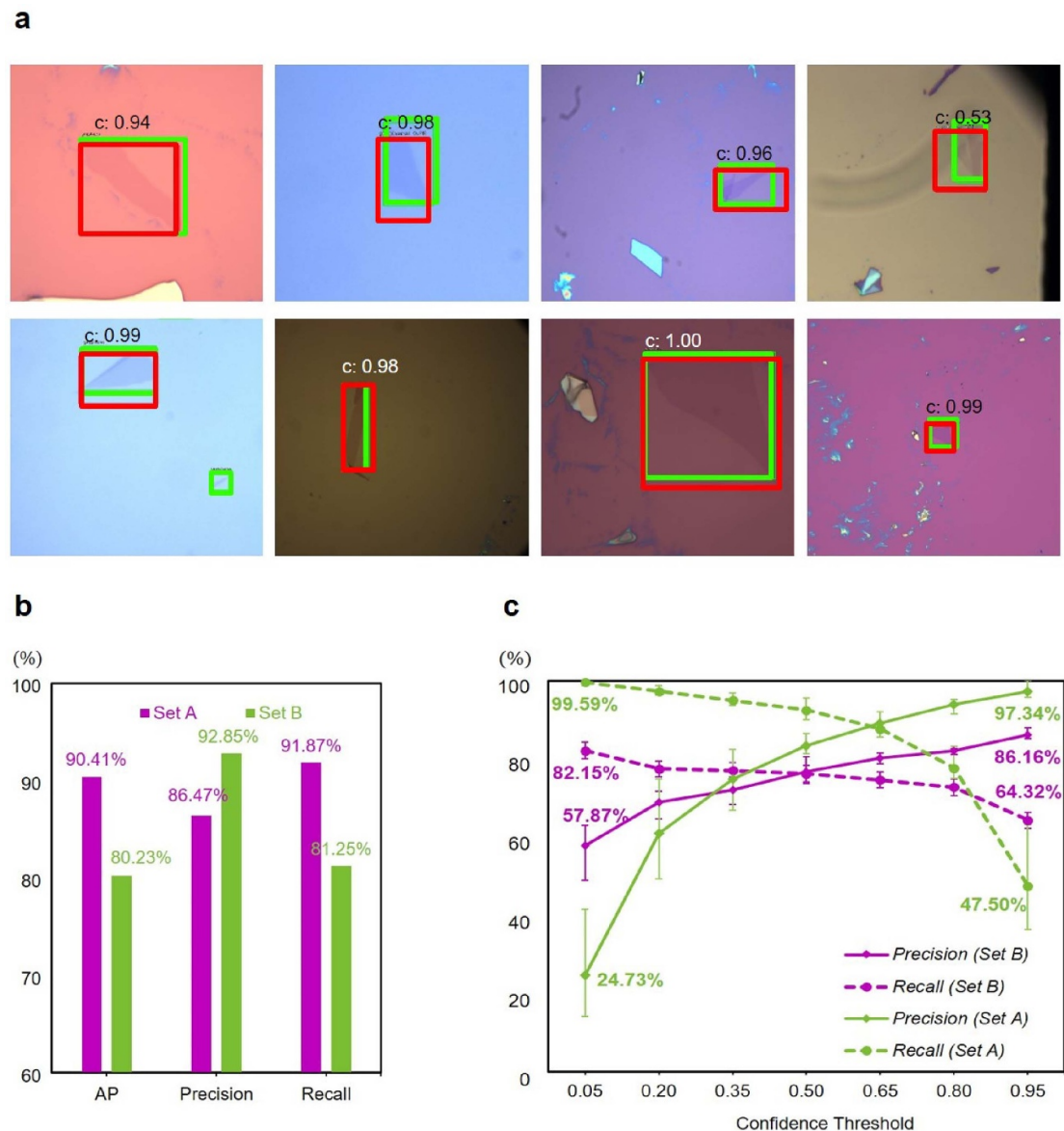


Figure 4. Detection results and performance graph. (a) The detection results (box and confidence(c)) on set G_{human} where the green boxes and red boxes in the figures are labeled by researchers and by the GT-DNN respectively. (b) The performance of our GT-DNN in terms of AP, precision and recall for two different target domains. (c) Performance evaluation, precision and recall curve under different confidence thresholds (IoU Threshold 0.3).

white balance. However, the scanning process creates other issues such as defocusing, unexpected junk flakes and scattered graphene because there is no human intervention during the process (figure S5).

The proposed fully automated GT-DNN achieved 80.23% in AP on Set G_{auto} (figure 4(b)), compared to 90.41% in AP on the Set G_{human} , likely due to a decreased Recall rate (81.25%) caused by tiny undetected graphene flakes under $10\ \mu m$ similar to debris. Error cases in Set G_{auto} are presented in figure S6 of the supplementary information section. Considering the Precision rate of 92.85%, which is higher than Precision on Set G_{human} , the GT-DNN shows comparable performance on Set G_{auto} .

For further exploration of the applicability for the proposed GT-DNN, we also train and evaluate the

GT-DNN on Set G_{auto} with a more detailed label set (mono-, bi- and tri-layered graphene). The experimental results show that our GT-DNN achieved 72.64% in AP for this multi-class detection task, which is lower than the 80.23% in AP for binary-class detection. This is mainly due to the fact that one graphene flake often contains different layered sub-graphene flakes while the label format for detection is limited to one label for each graphene flake. Therefore, the accuracy drop is likely caused by the duplicate boxes detected with multiple classes. The resulting cases are visualized in figure S6 of the supplementary information section.

To further enhance the applicability of the fully automated detection system, as shown in figure 4(c), we allow users to adjust the threshold by which users

can control the behavior of the GT-DNN in terms of the trade-off between precision and recall. For instance, high Precision may be preferred for industrial mass production in good quality while high Recall is more valued when searching for extremely rare flakes such as transition metal dichalcogenide monolayers or superconducting monolayers. Surprisingly, by adjusting the threshold, we found out that our GT-DNN can achieve remarkable performance in Precision (over 95% on Set G_{human}), which is a similar performance to trained researchers. In the next section, we will further explore how GT-DNN can help with advancing fundamental techniques in material science.

3.6. Finding best exfoliation methods and graphite crystal types using GT-DNN

In this section, we address a crucial issue in material science using our GT-DNN. Although there are multiple exfoliation methods that increase the yield of graphene flakes, the best exfoliation method remains unclear. Previously, the number of flakes and size were measured by humans [32, 33] and researchers preferred exfoliation methods and graphite crystal types based on their own experience. With our GT-DNN, we can count the number and size of graphene flakes automatically, which provides many samples that have statistical significance. In this section, using our GT-DNN, we first explore the best exfoliation methods and the best type of graphite crystals.

We compared three exfoliation methods: conventional 3 M tape, exfoliation with TRT [32] and a combination of both TRT and oxygen plasma treatment (UV ozone) [32] in terms of size and the number of flakes. And three types of graphite crystals: Kish Graphite (KishG) [33], NG [34] and HOPG [35] also in terms of size and the number of flakes. In the experiments, we used seven substrate chips for each exfoliation method and three chips for each graphite crystal. We used NG to test each exfoliation method and a combination of UV ozone and TRT to test each graphite crystals.

In table 1, the upper three rows compare exfoliation methods where the control is the conventional 3 M tape graphene exfoliation method. In terms of the mean top-5 sizes, there is not much difference between using TRT and using a combination of both TRT and oxygen plasma cleaning. According to the number of boxes detected by the GT-DNN, the combination method produces about 50% more graphene flakes in total. The combination method shows a slightly lower average box size than the TRT method because the combination method produces many smaller graphene flakes.

As shown in table 1, the bottom three rows compare graphite crystals. With the combination method, NG produces biggest graphene flakes in size. However, on average, KishG produces a greater number of graphene flakes. At this stage, this explains why

scientists prefer NG over other graphite crystals for vdW heterostructure assembly, because one or two big graphene flakes are necessary. However, for mass production, KishG may be a more suitable candidate. In this section, we prove that deep learning technology can contribute to material science by finding the best exfoliation method and type of graphite crystal.

3.7. Transferability to rare material detection

In this section, we will verify and investigate the transferability of the proposed GT-DNN to other materials. To train new deep learning models for other materials, a deep learning approach requires a lot of data for each material type, which is limiting. To reduce the amount of data required for new materials, we use transfer learning [36], which is an approach to utilizing knowledge from a model trained with previous data (pretrained model) to learn new data.

In general, the ImageNet [36] pretrained model has been adopted for transfer learning in many tasks, such as object detection [37], semantic segmentation [38] and action recognition [39]. However, in this paper, we conjecture that our GT-DNN (pretrained model with graphene data) provides higher feature representation power than the conventional ImageNet pretrained model when applied to vdW materials with similar textures and structural characteristics (graphene, hBN, etc).

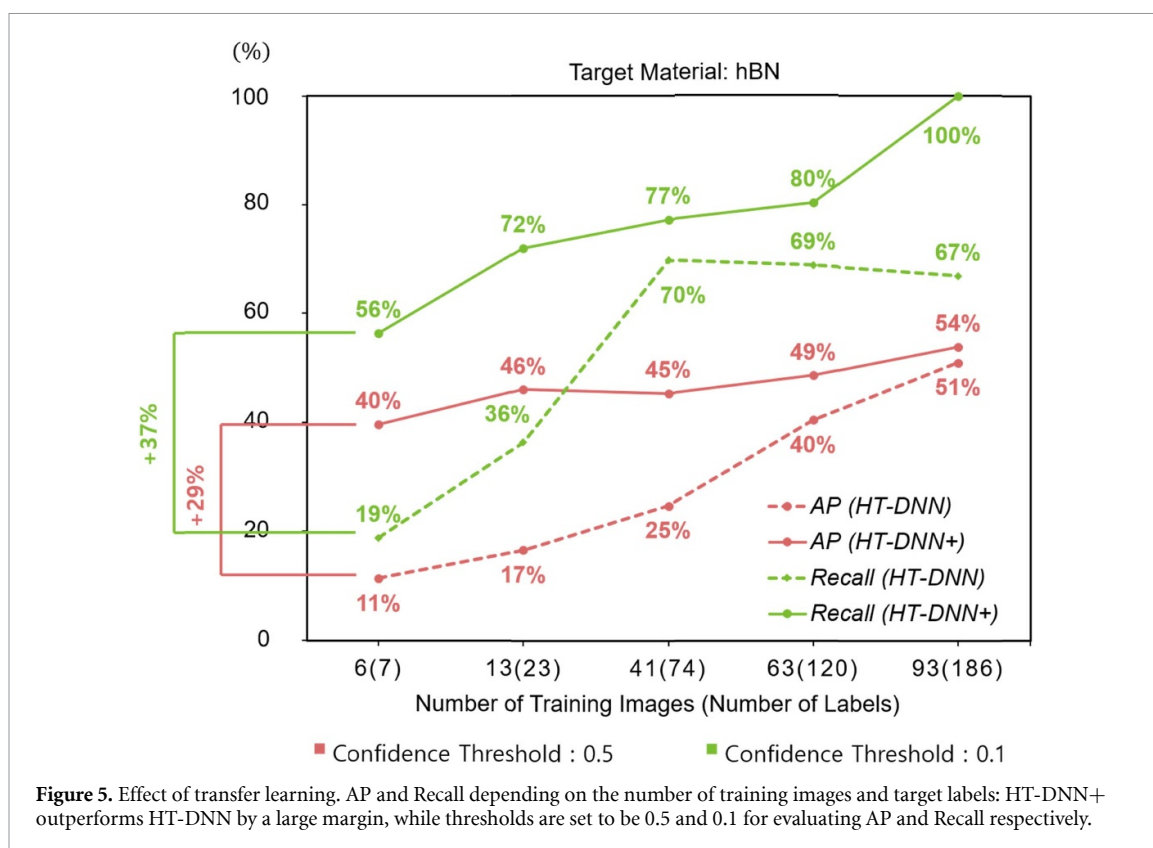
To verify the effectiveness of transfer learning with the GT-DNN for material detection, we train a detection model for hBN data (Set hBN_{auto}) transferred from our pretrained GT-DNN. For comparison, we train another detection model for Set hBN_{auto} transferred from the ImageNet pretrained DNN. We call this detection model as hBN trained DNN (HT-DNN) and especially we call the model transferred from GT-DNN as HT-DNN+.

In this experiment, for training networks, we vary the number of hBN training images in {6, 13, 41, 63, 93} of Set hBN_{auto} to see the effect of transfer learning depending on the number of training images. Figure 5 summarizes the results of AP and Recall for HT-DNN and HT-DNN+ on Set hBN_{auto} . As shown in figure 5, HT-DNN+ outperforms the HT-DNN by a large margin, especially when the number of samples is small (e.g. 6 and 13). Note that, in this experiment, we set the confidence threshold to 0.1, indicating that the network detects as many target flakes as possible. This is due to the fact that, for rare materials, missing target flakes is crucial while misclassifying them is less important.

Also, we found that HT-DNN does not increase Recall performance even if the number of hBN training samples increases, since the ImageNet pretrained model is not generalized for the shape of 2D material objects. This suggests the importance of using a proper pretrained model to train materials with extremely small datasets. Consequently, we prove for the first time that the proposed GT-DNN can be used

Table 1. Top-1 size, the average of the top-5 box sizes, the mean size of all boxes and the total number of boxes for three different exfoliation methods (conventional 3 M tape exfoliation [12], thermal releasing tape exfoliation and additional oxygen plasma treatment [32]) and three different graphite crystals (HOPG [35], natural graphite [34] and Kish graphite [33]).

Layers	Top-1 size (Mean top-5 size)			Mean size			Box no		
	Mono	Bi	Tri	Mono	Bi	Tri	Mono	Bi	Tri
Control	6.28 (4.12)	5.69 (3.94)	6.12 (4.35)	3.28	3.50	4.02	12.57	12.00	9.42
Thermal release	18.97 (14.21)	17.27 (13.41)	20.36 (16.69)	3.93	4.20	4.96	118.85	170.85	102.85
UV ozone and thermal release	14.98 (11.06)	17.52 (12.63)	21.85 (14.26)	3.89	3.98	4.75	335.57	408.42	253.14
HOPG	5.23 (3.22)	4.59 (2.04)	2.61 (1.60)	4.19	3.37	2.75	6.66	5.33	2.33
NG	9.18 (6.90)	9.52 (6.80)	11.43 (9.67)	3.32	3.39	3.97	80.33	95.33	93.00
KishG	6.69 (5.28)	7.81 (6.47)	10.68 (7.91)	2.97	2.98	3.48	154.50	164.00	103.00



as a baseline for other rarer materials with small training sets in future research. We will release the source code for the GT-DNN and HT-DNN+ for this purpose.

4. Conclusions

In this study, the automatic optical detection technique was developed with a DNN incorporated into a motorized microscope. The previous researches needed more human engagement. Using this technique graphene and thin hBN were detected and identified on silicon wafers automatically with high precision. Transfer learning helps develop thin hBN detection model with a few data set based on graphene data set. In near future, we like to expand the detection model trained by DNN to other rare 2D materials with low yield. So, scientist in 2D material area will get out of labor to find proper flakes.

Acknowledgments

Work by Young Jae Shin used resources at the Center for Functional Nanomaterials, which is a U.S. DOE Office of Science Facility, and at Brookhaven National Laboratory under Contract No. DE-SC0012704. This research was supported by the Korean MSIT (Ministry of Science and ICT), under the National Program for Excellence in SW (2017-0-00093), supervised by the IITP (Institute for Information & communications Technology Planning & Evaluation)

ORCID iDs

Young Jae Shin <https://orcid.org/0000-0002-7525-5062>

Kenji Watanabe <https://orcid.org/0000-0003-3701-8119>

Sung-Ho Bae <https://orcid.org/0000-0002-3389-1159>

References

- [1] Geim A K and Grigorieva I V 2013 Van der Waals heterostructures *Nature* **499** 419–25
- [2] Chakraborty C, Goodfellow K M, Dhara S, Yoshimura A, Meunier V and Vamvakas A N 2017 Quantum-confined Stark effect of individual defects in a van der Waals heterostructure *Nano Lett.* **17** 2253–8
- [3] Yi M and Shen Z 2015 A review on mechanical exfoliation for the scalable production of graphene *J. Mater. Chem. A* **3** 11700–15
- [4] Molina-Mendoza A J, Giovanelli E, Paz W S, Niño M A, Island J O, Evangeli C, Aballe L, Foerster M, Van Der Zant H S and Rubio-Bollinger G 2017 Franckeite as a naturally occurring van der Waals heterostructure *Nat. Commun.* **8** 1–9
- [5] Novoselov K S, Mishchenko A, Carvalho A and Neto A C 2016 2D materials and van der Waals heterostructures *Science* **353** 6298
- [6] Duong D L, Yun S J and Lee Y H 2017 van der Waals layered materials: opportunities and challenges *ACS Nano* **11** 11803–30
- [7] Ajayan P, Kim P and Banerjee K 2016 van der Waals materials *Phys. Today* **69** 38
- [8] Unuchek D, Ciarrocchi A, Avsar A, Watanabe K, Taniguchi T and Kis A 2018 Room-temperature electrical control of exciton flux in a van der Waals heterostructure *Nature* **560** 340–4

- [9] Ryu Y K, Frisenda R and Castellanos-Gomez A 2019 Superlattices based on van der Waals 2D materials *Chem. Commun.* **55** 11498–510
- [10] Cao Y, Fatemi V, Fang S, Watanabe K, Taniguchi T, Kaxiras E and Jarillo-Herrero P 2018 Unconventional superconductivity in magic-angle graphene superlattices *Nature* **556** 43–50
- [11] Yoo H, Engelke R, Carr S, Fang S, Zhang K, Cazeaux P, Sung S H, Hovden R, Tsen A W and Taniguchi T 2019 Atomic and electronic reconstruction at the van der Waals interface in twisted bilayer graphene *Nat. Mater.* **18** 448–53
- [12] Huang Y, Sutter E, Shi N N, Zheng J, Yang T, Englund D, Gao H-J and Sutter P 2015 Reliable exfoliation of large-area high-quality flakes of graphene and other two-dimensional materials *ACS Nano* **9** 10612–20
- [13] Masubuchi S, Morimoto M, Morikawa S, Onodera M, Asakawa Y, Watanabe K, Taniguchi T and Machida T 2018 Autonomous robotic searching and assembly of two-dimensional crystals to build van der Waals superlattices *Nat. Commun.* **9** 1–12
- [14] Goyal P and Kaiming H 2018 Focal loss for dense object detection *IEEE Trans. Pattern Anal. Mach. Intell.* **39** 2999–3007
- [15] Girshick R 2015 Fast R-CNN *IEEE Int. Conf. on Computer Vision (ICCV)* pp 1440–8
- [16] Ren S, He K, Girshick R and Sun J 2015 Faster R-CNN: towards real-time object detection with region proposal networks *Advances in Neural Information Processing Systems* pp 91–99
- [17] Lin T-Y, Dollár P, Girshick R, He K, Hariharan B and Belongie S 2017 Feature pyramid networks for object detection *Proc. of the IEEE Conf. on Computer Vision and Pattern Recognition* pp 2117–25
- [18] Liu W, Anguelov D, Erhan D, Szegedy C, Reed S, Fu C-Y and Berg A C 2016 SSD: single shot multibox detector *European Conf. on Computer Vision (Springer)* pp 21–37
- [19] Li Y, Chen Y, Wang N and Zhang Z 2019 Scale-aware trident networks for object detection *Proc. of the IEEE Int. Conf. on Computer Vision* pp 6054–63
- [20] Han B, Lin Y, Yang Y, Mao N, Li W, Wang H, Fatemi V, Zhou L, Wang J I-J and Ma Q 2019 Deep learning enabled fast optical characterization of two-dimensional materials (arXiv:1906.11220)
- [21] Saito Y, Shin K, Terayama K, Desai S, Onga M, Nakagawa Y, Itahashi Y M, Iwasa Y, Yamada M and Tsuda K 2019 Deep-learning-based quality filtering of mechanically exfoliated 2D crystals *npj Comput. Mater.* **5** 1–6
- [22] He K, Zhang X, Ren S and Sun J 2016 Deep residual learning for image recognition *IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)* pp 770–8
- [23] Kingma D and Ba J 2014 Adam: a method for stochastic optimization *Int. Conf. Learn. Represent*
- [24] He K, Girshick R and Dollar P 2019 Rethinking imagenet pre-training *IEEE/CVF Int. Conf. on Computer Vision (ICCV)* pp 4917–26
- [25] Shorten C and Khoshgoftaar T M 2019 A survey on image data augmentation for deep learning *J. Big Data* **6** 60
- [26] Chollet F 2018 *Keras: The Python Deep Learning Library* (Astrophysics Source Code Library)
- [27] Ketkar N 2017 *Deep Learning with Python* (Berkeley, CA: Apress) pp 195–208
- [28] Abadi M et al 2016 TensorFlow: a system for large-scale machine learning *Proc. 12th USENIX Symp. on Operating Systems Design and Implementation (Savannah, GA)* (Savannah, GA: USENIX Association) pp 265–84
- [29] Liu L, Ouyang W, Wang X, Fieguth P, Chen J, Liu X and Pietikäinen M 2020 Deep learning for generic object detection: a survey *Int. J. Comput. Vis.* **128** 261–318
- [30] Maaten L V D and Hinton G 2008 Visualizing data using t-SNE *J. Mach. Learn. Res.* **9** 2579–605
- [31] Selvaraju R R, Cogswell M, Das A, Vedantam R, Parikh D and Batra D 2017 Grad-CAM: visual explanations from deep networks via gradient-based localization *IEEE Int. Conf. on Computer Vision (ICCV)* pp 618–26
- [32] Pizzocchero F, Gammelgaard L, Jessen B S, Caridad J M, Wang L, Hone J, Bøggild P and Booth T J 2016 The hot pick-up technique for batch assembly of van der Waals heterostructures *Nat. Commun.* **7** 1–10
- [33] Morita S, Tsukada S and Mikoshiba N 1988 Scanning tunneling microscopy of Kish graphite and highly oriented pyrolytic graphite in air *J. Vac. Sci. Technol.* **6** 354–7
- [34] Wang X, Fulvio P F, Baker G A, Veith G M, Unocic R R, Mahurin S M, Chi M and Dai S 2010 Direct exfoliation of natural graphite into micrometre size few layers graphene sheets using ionic liquids *Chem. Commun.* **46** 4487–9
- [35] Chung D D L 1987 Exfoliation of graphite *J. Mater. Sci.* **22** 4190–8
- [36] Russakovsky O, Deng J, Su H, Krause J, Satheesh S, Ma S, Huang Z, Karpathy A, Khosla A and Bernstein M 2015 Imagenet large scale visual recognition challenge *Int. J. Comput. Vis.* **115** 211–52
- [37] Girshick R, Donahue J, Darrell T and Malik J 2014 Rich feature hierarchies for accurate object detection and semantic segmentation *IEEE Conf. on Computer Vision and Pattern Recognition* pp 580–7
- [38] Shelhamer E, Long J and Darrell T 2017 Fully convolutional networks for semantic segmentation *IEEE Trans. Pattern Anal. Mach. Intell.* **39** 640
- [39] Simonyan K and Zisserman A 2014 Two-stream convolutional networks for action recognition in videos *Advances in Neural Information Processing Systems* pp 568–76